GPS Spoofing Detection on Autonomous Vehicles with XGBoost

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***Abstract*—Global Positioning System (GPS) spoofing presents a significant threat to the reliability and security of global navigation systems by impacting various critical sectors such as military, transportation, communication, and finance. Traditional methods for detecting GPS spoofing often fall short in addressing sophisticated spoofing attacks. In our work, we investigate the application of machine learning techniques to enhance the detection of GPS spoofing. We aim to identify patterns and anomalies in GPS signals that indicate spoofing attempts by leveraging data-driven approaches. Our proposed method involves training machine learning models on a dataset comprising both legitimate and spoofed GPS signals. The results obtained from this work demonstrate the effectiveness of these models in accurately detecting spoofing incidents, particularly with the tree-based machine learning model XGBoost. This research underscores the potential of machine learning to provide robust, real-time spoofing detection, thereby enhancing the resilience of GPS-dependent systems against malicious attacks, with XGBoost standing out for its high accuracy.**

Keywords*—GPS spoofing detection, machine learning, cybersecurity, autonomous vehicles

I. Introduction

Global Positioning System (GPS) technology has become an integral part of modern infrastructure, providing essential services in navigation, timing, and location-based applications across various industries. From autonomous vehicles to financial systems, the reliance on accurate GPS signals is perva-sive and critical, underpinning everything from transportation networks and emergency services to military operations and financial transactions. However, this widespread dependence also introduces significant vulnerabilities. Among these is GPS spoofing, a form of cyber-attack where false GPS signals are deliberately transmitted to deceive GPS receivers. This can lead to incorrect position, velocity, and time information, potentially causing severe disruptions. GPS spoofing presents a substantial risk to the integrity and reliability of navigation and timing systems. Traditional detection methods, such as signal strength monitoring and anomaly detection based on signal properties, are inadequate against sophisticated spoofing attacks that can closely mimic legitimate GPS signals. These advanced spoofing techniques can disrupt navigation systems, misguide autonomous vehicles, interfere with the timing of financial transactions, and compromise the operations of critical infrastructure. Given the evolving nature of spoofing threats, there is acrucial need for more robust and adaptive detection mechanisms. Recent advancements in machine learning offer promising ways for enhancing GPS spoofing detection. Machine learning algorithms can analyze large volumes of data to identify subtle patterns and anomalies that are indicative of spoofing. By leveraging these data-driven approaches, it is possible to develop models that can accurately distinguish between legitimate and spoofed GPS signals in real-time.

In this work, we explored the application of machine learning techniques to address the challenge of GPS spoofing detection in autonomous vehicles. We propose a novel approach that combines feature preprocessing with supervised learning algorithms to effectively identify spoofing attempts. Our methodology involves the careful engineering of features from a comprehensive dataset containing both legitimate and spoofed GPS signals. This dataset is then used to train and evaluate various machine learning models. The focus of our work is on optimizing the feature selection process, refining the model training phase, and thoroughly assessing the detection performance of our proposed model. Through rigorous experimentation and analysis, we demonstrate the effectiveness of our approach in enhancing the security and reliability of GPS-based navigation in autonomous vehicles.

By leveraging the power of machine learning, we aim to enhance the resilience of GPS-dependent systems against spoofing attacks, ensuring greater security and reliability in critical applications.

The remainder of this paper is organized as follows: Section II reviews GPS spoofing detection with signal analysing, machine learning applications and deep learning based solutions. Section III describes a new machine learning-based GPS spoofing detection approach using XGBoost, including data preprocessing, model training, testing and evaluation of results. Section IV presents the experimental evaluation and dataset, demonstrating the effectiveness of our approach. Finally, Section V conclude the implications of our findings and outlines future research directions.

II. GPS Spoofing and Machine Learning

A. Signal Analysis Based Researches

Feng et al. proposed a two-pronged approach to detect GPS spoofing on UAVs. The first method involved comparing the calculated acceleration from gyroscope and accelerometer data with the GPS-provided position data to identify discrepancies that could indicate spoofing [1]. The second method compared yaw angles derived from gyroscope data with those calculated from GPS data, where significant differences could suggest

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tampered GPS information [2]. Keum and Duk developed a simpler method that directly compared the acceleration estimated from GPS data with that measured by the UAV's accelerometer. Discrepancies between these values would signal potential GPS spoofing, as the GPS-derived acceleration would deviate from the actual acceleration recorded by the accelerometer [3].

B. Machine Learning Based Researches

Meng et al. introduced an anti-spoofing model based on Linear Regression (LR) that is specifically developed to predict and map out the ideal route for a UAV. This model is effective in both identifying GPS spoofing attempts and reducing their effects, thereby allowing the UAV to carry on with its mission securely, even during an attack. It utilizes linear regression (LR) to predict longitude and latitude, allowing for comparison with the GPS-derived position. The performance of this approach was evaluated through simulation. However, like the methods proposed by other researchers, this approach has limitations as it relies on only a few types of sensor data. The study also compares the LR model to alternative approaches like Long Short-Term Memory (LSTM) networks, demonstrating that the LR model achieves superior prediction accuracy [4].

Shafique, Mehmood and Elhadef introduced a machine learning-driven approach for detecting GPS signal spoofing attacks targeting Unmanned Aerial Vehicles (UAVs). It primarily employs Support Vector Machine (SVM) as the main algorithm, accompanied by an in-depth evaluation of various machine learning models to identify the most effective one. The authors created multiple K-learning models using K-fold cross-validation, which enhances detection accuracy through a voting process. The system authenticates incoming GPS signals by analyzing specific signal characteristics, such as jitter and shimmer metrics. The UAV is configured to respond only to verified GPS signals, thereby reducing the risk of spoofing attacks. This method is optimized for computational efficiency, making it ideal for real-time UAV applications [5].

According to Filippou et al., GPS location spoofing attacks as anomalies and have developed a machine learning-based anomaly detection solution to identify them. Notably, this solution does not require any attack data during the training phase and still achieves impressive classification accuracy, with F1 and G_m scores ranging from 95% to 98%, depending on the machine learning model used. Among the techniques tested, the variant using the local outlier factor (LOF) model demonstrated the highest detection accuracy, surpassing all other ML-based approaches and outperforming a non-ML state-of-the-art solution by 15% [6].

Kim et al. proposed a detection method using a multilayer perceptron (MLP) model trained on data from the gyroscope, accelerometer, and GPS. The method's performance was assessed using a software-in-the-loop (SITL) simulation. While these methods incorporate machine learning algorithms to analyze perception data, they only utilize a limited range of sensor data. This approach overlooks the inherent limitations of sensors, such as vibration effects on the accelerometer, cumulative errors in the gyroscope, electromagnetic interference affecting the magnetometer, and wind impact on the barometer. As a result, sensor errors could be mistakenly identified as spoofing attacks [7].

Feng et al. introduced a GPS spoofing detection method for UAVs using the XGBoost model, a highly efficient and accurate gradient boosting machine learning technique. The model's features included angular velocity, acceleration, and distance, with the distance calculated based on changes in longitude and latitude between successive time points. This binary classification method allowed for the rapid and accurate identification of GPS spoofing attacks on UAVs. By focusing on these specific features, the approach was designed to detect anomalies in the UAV's movement that signal potential spoofing. XGBoost's application resulted in a robust model capable of managing the complexities of real-time UAV navigation data. This method offers a reliable and scalable way to enhance UAV security, particularly in scenarios where GPS spoofing presents significant threats [8].

Panice et al. proposed a detection method for UAV GPS spoofing attacks using a Support Vector Machine (SVM) model. The approach was centered on identifying discrepancies between the positions determined by GPS signals and those measured by the inertial navigation system (INS). Specifically, the SVM model was trained on the error values generated from these positional differences, enabling the system to detect inconsistencies that could indicate a spoofing attack. The dataset used for training the SVM model was obtained through simulation, ensuring that a variety of scenarios could be modeled to enhance the robustness of the detection method. This technique capitalizes on the complementary strengths of GPS and INS data, allowing the SVM to effectively distinguish between normal operational variations and potential spoofing attempts. The use of SVM, a well-established machine learning algorithm known for its accuracy in classification tasks, provided a reliable means of enhancing the security of UAV operations by detecting GPS spoofing attacks with precision [9].

Wei et al. introduced the perception-data-based UAV GPS spoofing detection (PerDet) approach, a machine-learning-based method for detecting GPS spoofing in UAVs using perception data. Their approach involves a thorough analysis of the UAV's perception data through both the position and attitude estimation processes. We utilize data from the accelerometer, gyroscope, magnetometer, GPS, and barometer as features, allowing these sensors to complement and verify each other's measurements. The key distinction between PerDet and existing methods lies in our selection process for perception data. Additionally, unlike many existing detection methods that are validated through simulations, we collect our dataset from real flight experiments [10].

C. Deep Learning Based Researches

Calvo-Palomino et al. developed a GPS spoofing detection method by training a Long Short-Term Memory (LSTM) model using GPS Doppler shift measurements at 5-second intervals. The LSTM model, known for its ability to handle time-series data, was used to predict patterns in the Doppler effect, which reflects the UAV's movement relative to GPS satellites. By comparing the observed Doppler shift with the model's predictions, the system could identify discrepancies indicative of spoofing. Simulation experiments validated the method's effectiveness, demonstrating its potential to enhance GPS-based system security, especially in UAVs [11].

In data driven learning models, first and significant part of work is collecting data and departing from dataset unnecessary

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records. Sun et al., used Spearman's rank correlation coefficient to preprocess the GPS signal dataset to remove irrelevant and weakly correlated data. They utilized the SVM-SMOTE method for data augmentation in order to enhance the training and generalization capabilities of the deep learning model. They proposed a solution to train model as PCA-CNN-LSTM. PCA comes from principal component analysis is a method to preprocess the dataset, Convolutional Neural Network (CNN) is the model to make sense of dataset and long short-term memory (LSTM) is used as submodule of CNN to further process for model [12].

Korium et al. proposed an image-based intrusion detection system for detecting GPS spoofing attacks in UAVs using a deep learning approach that combines CNNs with PCA for dimensionality reduction, data augmentation to enhance the training dataset, and transfer learning to improve performance with limited data. This method aims to provide a fast, accurate, and general solution for detecting cyberattacks. Extensive numerical experiments demonstrated its high accuracy and low latency, outperforming existing models like Inception-v3. The approach effectively addresses UAV vulnerabilities to GPS spoofing, making it a robust tool for enhancing UAV cybersecurity in both military and civilian applications [13].

Wang et al. developed a GPS spoofing detection method using a LSTM network. Their training data included velocity, acceleration, and positional coordinates (latitude and longitude) along the x and y axes of UAVs. The LSTM model was trained to predict the UAV's position based on its flight path and the provided feature data. For detection, the UAV was required to follow a predefined path, and the method worked by comparing the predicted position with the actual position reported by the GPS receiver. The difference between these positions was measured against a specified threshold to determine the presence of a GPS spoofing attack [14].

Dang et al., presented a UAV monitoring and anti-GPS spoofing system that utilizes cellular connectivity. The system uses path loss measurements from base stations to verify the UAV's GPS coordinates. It combines statistical methods with deep learning, specifically employing a Multi-Layer Perceptron (MLP) model, to create a novel GPS spoofing detection approach. The results show that this approach is highly effective, achieving over 93% accuracy with data from three base stations and up to 80% accuracy with just one base station, thanks to the stability provided by the statistical metrics [15].

Borhani-Darian et al. explored deep neural networks for detecting GNSS spoofing attacks, building on prior research that used deep learning to differentiate legitimate GNSS signals from noise. The study highlights the complexity of spoofing detection, making deep learning models essential. They introduced a data-driven classifier using an image-splitting process for parallelization and a Gaussian mixture model to estimate the number of spoofed signals. This approach is more computationally intensive than standard GNSS detection methods, requiring multiple neural network models. While parallelization improves performance, further optimization is needed for high-rate detection scenarios. The results show that this deep learning method outperforms existing approaches, especially at moderate to high signal-to-noise ratios, but may still struggle to differentiate between multipath and spoofing signals. Future research is expected to focus on distinguishing spoofing from multipath effects, particularly in environments like urban canyons [16].

Bose proposed a neural network-based anti-spoofing method that can be added to existing GPS receivers as a software upgrade, avoiding the need for significant hardware or software changes. This supervised machine learning approach uses accessible GPS observables, like pseudorange and Doppler shift, to distinguish between real and spoofed signals. The neural network's architecture, training, and testing details were discussed, and simulations showed that the method can detect spoofing signals with 99% accuracy and only a 1% chance of misclassification [17].

III. A New Approach for GPS Spoofing Detection with XGBoost

In our work, we explore GPS spoofing detection using machine learning methods. Initially, we conducted a literature review to identify a suitable dataset for our research. We discovered the AV-GPS-Dataset [18], which aligns well with our objectives. Using this dataset, we trained various machine learning models to detect GPS spoofing attacks. We then thoroughly evaluated the performance of these models, analyzing their effectiveness in identifying spoofing attempts based on the provided data. In our observatios, we get a better accuracy with XGBoost, so we based on this model. In Figure 1, all steps of process can be shown. After obtaining a dataset related to our task, investigation of features and modifying dataset phase comes. We explain them in preprocessing data section. Then, spliting a dataset into 10-fold phase should be applied to obtain suitable result from model. Last two steps are explained in training and testing models section.

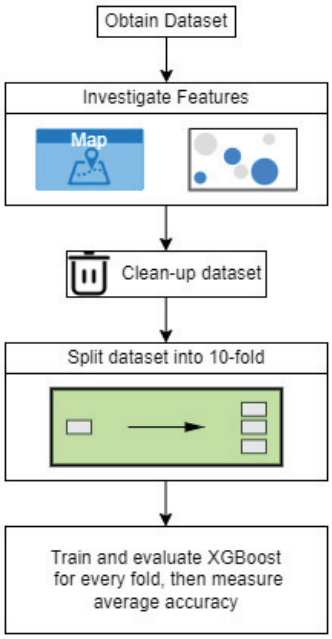

Fig. 1. Workflow of process

A. Preprocessing Data

The first step in our work is feature engineering, which involves identifying and selecting the relevant features from the dataset. This process includes removing irrelevant or redundant fields and handling missing data to ensure the dataset is clean and complete. By refining the features and addressing data quality issues, we aim to enhance the accuracy of the learning process, enabling the model to learn more effectively from the

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data. This step is crucial for improving the performance of the machine-learning algorithms and achieving reliable results in detecting GPS spoofing attacks.

The dataset comprises 44 columns, including one label column, and contains a total of 62,042 samples. Some columns in the dataset include non-numeric values that are not relevant for determining whether a GPS signal is normal or compromised. Therefore, these non-essential features are removed from the dataset before the model training phase. This preprocessing step is crucial to ensure that the data used for training the machine learning models is both meaningful and suitable for accurately identifying GPS spoofing signals. By eliminating irrelevant features, we improve the quality of the dataset and enhance the performance of the models in detecting spoofing attacks. Below features are removed form dataset before model training phase.

- Run Time
- Hobbs
- Clock Time
- Clock Date
- GPS MGRS

Some samples in the dataset have missing values in their columns. To address this issue, we fill the missing values with the mean of the respective columns. This approach helps maintain the integrity of the dataset and ensures that the missing values do not negatively impact the model's performance. Additionally, before the training phase, all features are normalized to standardize the data. Standard score of a sample x is calculated as in Equation 1 where u is the mean of the training samples and s is the standard deviation of the training samples. Normalization adjusts the scale of the features to a common range, which helps improve the accuracy and effectiveness of the machine learning model by ensuring that no single feature disproportionately influences the model's results. This preprocessing step is essential for achieving more reliable and consistent outcomes from the model.

$$z = (x - u)/s \tag{1}$$

B. Training and Testing Models

Before the training and testing stages, the dataset needs to be split into separate training and testing subsets. This process involves partitioning the data so that a portion is used to train the machine learning model, while the remaining portion is reserved for evaluating the model's performance. Splitting the dataset ensures that the model can be trained on one set of data and tested on an independent set, providing an unbiased assessment of its accuracy and generalizability. This separation helps to prevent overfitting and ensures that the model's performance metrics reflect its ability to make predictions on new, unseen data. We used 10-fold cross-validation to split the dataset into training and testing subsets. As seen in Figure 2 this technique divides the dataset into ten equally sized folds. For each iteration, nine folds are used for training the model, while the remaining fold serves as the testing dataset. This process is repeated ten times, with each fold being used as the test set once. As a result, we obtain ten different training and testing sets, allowing us to assess the model's performance from multiple perspectives.

The final performance metrics are averaged across these ten iterations to provide a more reliable and robust evaluation of

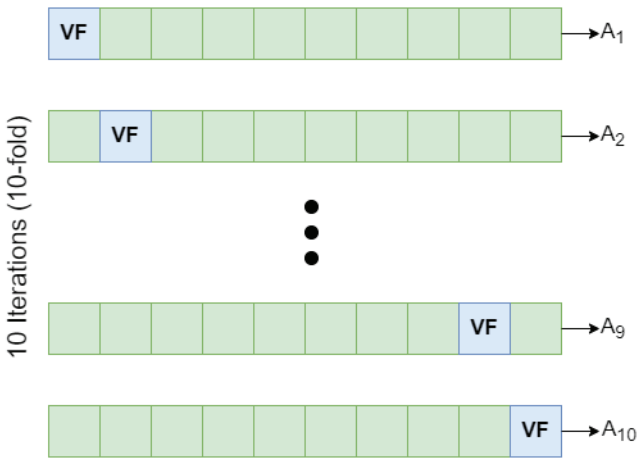

Fig. 2. 10-fold cross validation

the model's effectiveness and applicability shown in Equation 2.

This approach helps to mitigate the impact of any single train-test split and ensures that the model is validated on various subsets of the data.

$$A = \frac{1}{10}\sum_{i=1}^{10} A_i \tag{2}$$

IV. EXPERIMENTAL EVALUATION

A. Development Environment

We used Google CoLab as a development environment. It obtains a ready setup to run machine learning algorithms without any installation process. Python programming language is used in development. Below Python libraries are used.

- sklearn
- xgboost
- pandas
- folium

Hardware specifications of development environment can be shown in Table I.

TABLE I HARDWARE SPECIFICATIONS

Parameter	Specification
Disk Space	100 GB
Available RAM	12 GB
Number of CPU	2
CPU Model Name	Intel(R) Xeon(R)
CPU Frequency	2.20GHz
GPU Model Name	NVIDIA Tesla K80
GPU Memory	12 GB

B. Dataset

Used dataset consists of three subset with different characteristics [18], created by Abrar et al. [19]. Three dataset named as AV-GPS-Dataset-1, AV-GPS-Dataset-2 andAV-GPS-Dataset-3. In our work only AV-GPS-Dataset-1 is used. Main difference between datasets is GPS data location. Collection of GPS data is processed in different locations.

In AV-GPS-Dataset-1, there are 62043 samples and 43 features. An additional column in dataset named as "Data Type" is label of the sample. It has value "0" for normal GPS data and "1" for malicious data. In this dataset 46287 normal

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'Roll (deg)'	'Steering Angle (deg)'	'Longitudinal Velocity (m/s)'
'Pitch (deg)'	'Mission Index'	'Heading To Next WP (deg)'
'Yaw (deg)'	'Vertical Position (m)'	'Altitude Tuning Setpoint'
'GPS Latitude'	'Relative Altitude (m)'	'Altitude AMSL (m)'
'Throttle (%)'	'GPS Longitude'	'Longitudinal Position (m)'
'Altitude Tuning'	'Lateral Velocity (m/s)'	'Absolute Longitudinal Velocity (m/s)'
'GPS VDOP'	'Distance To Home (m)'	'X-Track Error (m)'
'Heading (deg)'	'Travelled Distance (m)'	'Satellite Locks'
'GPS HDOP'	'Temperature (F)'	'Longitudinal Vibration'
'Satellite Count'	'Vertical Velocity (m/s)'	'Distance To GCS (m)'
'Lateral Vibration'	'Lateral Position (m)'	'Heading To Home (deg)'
'GPS Course'	'Vertical Vibration'	'Velocity (m/s)'
'Absolute Lateral Velocity (m/s)'		'Yaw Rate (deg/s)'

Fig. 3. Used features of GPS dataset

GPS data, 15756 malicious GPS data exist. Some features are irrelevant for model training stage, so 38 features are used to train a models. In dataset, there are some blank fields, thus we deal with this problems by filling with mean value of each feature.

Used features are shown in Figure 3 with their units.

We investigate the GPS coordinates of each class on the map to understand pattern of movement by using Python Folium library. We marked coordinates of normal data with blue and coordinates of malicious data with red marker as in Figure 4. This observation obtains which locations are in our consideration and awareness of how GPS signal changing its trend.

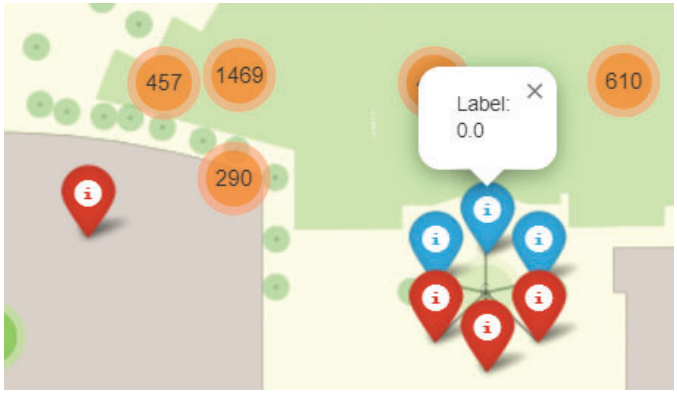

Fig. 4. Marked coordinates of GPS dataset

C. Evaluation of Results

The results of the machine learning models are presented in Table II. The models listed in the table were trained and tested using 10-fold cross-validation. As a result, the average performance across the ten iterations was considered as the final performance metric for each model. According to the results, the XGBoost model achieved the highest accuracy, demonstrating its superior performance in detecting GPS spoofing attacks. The decision tree model also performed well, securing the second-highest accuracy. Both XGBoost and decision tree models are tree-based machine learning algorithms, which explains their similar behavior in terms of classification accuracy. These models are particularly effective in handling structured data and capturing complex patterns, contributing to their strong performance in this application. The consistent results across these tree-based models suggest that they are well-suited for this type of classification task, offering both reliability and precision.

TABLE II RESULTS OF ML MODELS

	Mean Accuracy	Std of CV Accuracy
Decision Tree	0.97466	0.06983
Naive Bayes	0.97228	0.05124
MLP	0.95376	0.08050
KNN	0.94917	0.09725
XGBoost	0.97644	0.06625
SVM	0.94665	0.1034
LR	0.94072	0.09931
Random Forest	0.97140	0.06341

Second column standard deviation of cross validation accuracy indicates that variability or dispersion of the model's performance across different folds during the cross-validation process, thus smaller value is reason for preference after consideration of accuracy value.

Figure 5 show clearly accuracy comparison of ML models. Naive Bayesian classifier has the worst accuracy while XGBoost has the best accuracy result.

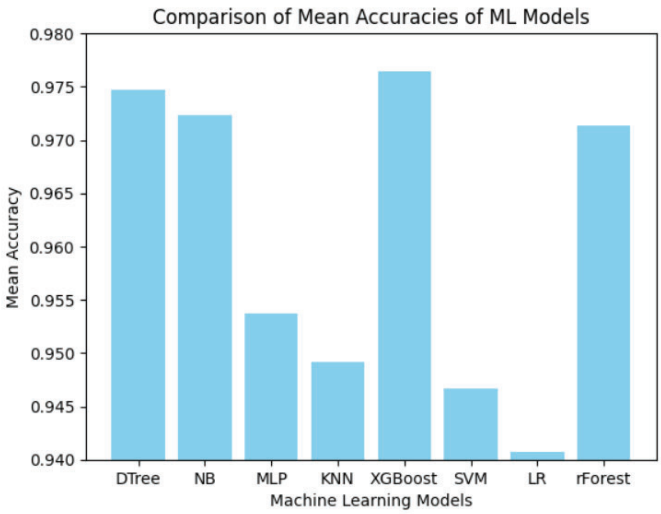

Fig. 5. Graphical representation of results

V. CONCLUSION

In conclusion, our investigation into the application of machine learning for GPS spoofing d etection h as shown promising results, particularly in the use of the XGBoost model for accurately identifying spoofing i ncidents. A n accuracy 0.976% is obtained with XGBoost which better than other ML models. XGBoost get accurate result because its characteristics compatible with sophisticated data. It can model complex and non-linear relationships in dataset. The success of this approach highlights the significant p otential o f data-driven methods in fortifying GPS-dependent systems against sophisticated spoofing a ttacks. B y e ffectively distinguishing between legitimate and malicious signals, our work contributes to enhancing the security and reliability of critical sectors that rely heavily on GPS technology. As GPS spoofing threats continue to evolve, the integration of advanced machine learning techniques, like those demonstrated in this study, will be crucial in developing resilient, real-time detection mechanisms that can adapt to and mitigate emerging challenges.

In future work, we plan to explore the incorporation of behavioral analysis techniques to enhance the detection of GPS spoofing. B y e xamining t he p atterns a nd a nomalies in

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GPS location data over time, we aim to develop models that can not only detect spoofing a ttempts b ut a lso identify subtle behavioral shifts indicative of more sophisticated attacks. This approach could provide an additional layer of defense by understanding the normal movement patterns of GPS-enabled devices and recognizing deviations that suggest malicious activity. Such advancements would further improve the robustness and adaptability of GPS spoofing detection systems.

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